1. Identity and decision problem

Mohammed Osama Hasan, registration number 25/27014, username ug2527014@ines.ac.rw. Verification code AI-A2-7014. Course: SWE 3513 Artificial Intelligence, INES-Ruhengeri. System commit 05c06d9a76a95bf3bdb8fde4d824ad8ba50c3825 on branch dataset-generation-validation.

Stakeholder and questions

The stakeholder is a field officer for a potato cooperative in Musanze. Farm records arrive in inconsistent spreadsheets, and the officer needs a local tool that can answer three questions before a visit.

- How many tons is this farm likely to harvest? This is the regression task, target yield_tons.
- Is the farm at high production risk? high_risk is 1 when yield per hectare on the generated base rows is below the median. This is the classification task.
- Which farms share similar growing conditions? This is the clustering task.

Ethical boundary

This is a classroom decision-support prototype. The dataset is synthetic. It is generated from registration number 25/27014 and it is not a record of real Musanze farms. The service does not replace an agronomist. It must not be used to grant or deny finance, insurance, subsidies, or any public service. A person reviews every estimate before anyone acts on it.

2. Data and reproducibility

LLLL is 7014, the last four digits of 25/27014. The generator calls numpy.random.default_rng(7014) and rejects any other seed. Base rows = 240 + (7014 mod 41) = 243. m = round(0.05 x 243) = 12, so 12 soil-moisture cells and 12 fertilizer cells are cleared. Three exact duplicate rows are then appended. The saved file has 246 rows. Numeric values use four decimal places.

SHA-256 of the CSV bytes, also stored in artifacts/data_fingerprint.txt:
209caa6bff4616bdeee0e6de48554e870d27289974057935d7287f4655323b10

Column	Rule	Role
rainfall_mm	Uniform 600 to 1400	Feature
avg_temp_c	Uniform 12 to 26	Feature
soil_moisture_pct	Uniform 20 to 80, then 12 missing	Feature
fertilizer_kg	Uniform 0 to 250, then 12 missing	Feature
field_size_ha	Uniform 0.2 to 4.0	Feature
season	A or B	Feature
cooperative_support	0 or 1	Feature
yield_tons	Formula below, minimum 0.50	Regression target
high_risk	1 if yield per hectare is below the median	Classification target

yield_tons = 0.004 x rainfall + 0.06 x soil moisture + 0.02 x fertilizer + 1.30 x field size - 0.25 x absolute value of (temperature - 18) + 0.70 x cooperative support + season effect + noise. Season A adds 0.50 and season B subtracts 0.20. Noise is normal with mean 0 and standard deviation 0.80. The result is clipped at 0.50 tons. high_risk is computed on the complete base rows, before the missing cells are planted.

Validation checks the column names, the ranges, the 12 missing values on the deduplicated rows, and the 3 extra copies. Training then drops those 3 copies before the split, so one farm cannot appear in both sets. Moisture and fertilizer stay missing until the imputer, and that imputer is fit on the training rows only.

3. Pipeline design

After the copies are removed, 243 farms remain. train_test_split uses test_size 0.2, random_state 7014, and stratifies on high_risk. That gives 194 training farms and 49 test farms. Regression, classification, and clustering all use this same split.

Train-only preprocessing

Each model is a scikit-learn Pipeline. The first step is a ColumnTransformer. Numeric columns are rainfall, temperature, soil moisture, fertilizer, field size, and cooperative support. They pass through a median imputer and then a standard scaler. Season passes through a most-frequent imputer and then a one-hot encoder. fit is called on the training frame only. The test frame is transformed with that fit.

Leakage controls

- Inputs are the seven growing-condition columns. yield_tons, yield per hectare, and high_risk are rejected as inputs.
- high_risk is not used to predict yield, and yield is not used to predict high_risk.
- Cluster profiles describe growing conditions. They do not use the targets.
- Duplicate rows are removed before the split, so a copied farm cannot leak into the test set.

Candidates

- Regression: a mean baseline, linear regression, and a random forest of 200 trees with max depth 6, min_samples_leaf 3, and random_state 7014.
- Classification: a majority-class baseline, logistic regression, and a random forest with the same forest settings. The positive class is high_risk = 1.
- Clustering: KMeans for K = 2, 3, 4, 5, and 6, with n_init 10 and random_state 7014. K is the value with the highest training silhouette.

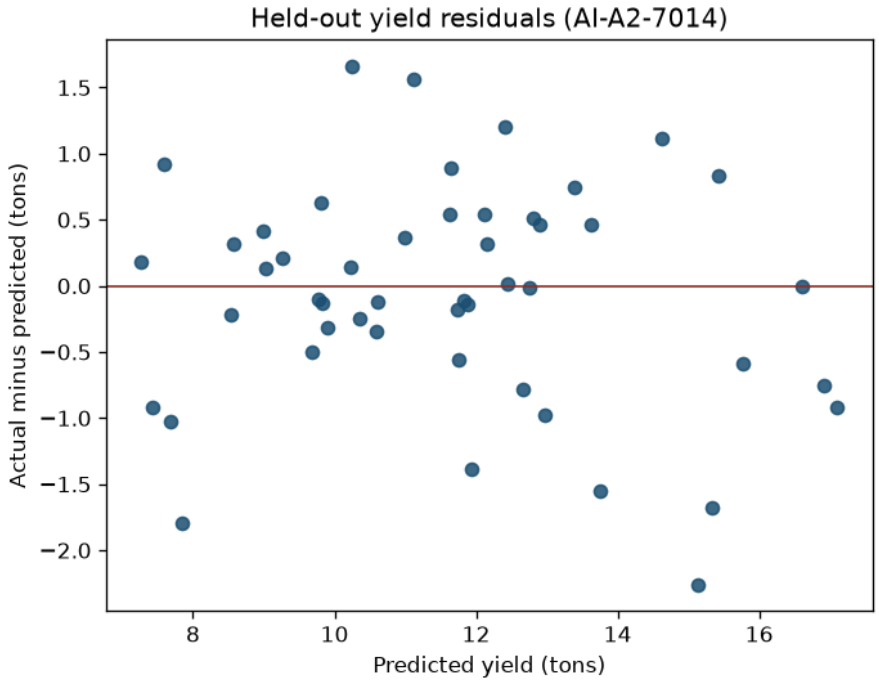
4. Regression and classification

Yield, held-out test set

Model	MAE	RMSE	R2
mean_baseline	2.058911	2.600897	-0.029191
linear_regression	0.649617	0.842947	0.891894
random_forest	1.021425	1.336125	0.728391

linear_regression has the lowest held-out RMSE (0.8429 tons), compared with the mean baseline RMSE of 2.6009 tons. These figures are the values stored in artifacts/regression_metrics.json.

The residual plot shows test errors scattered around zero across the predicted range. I do not see a separate band of failures at only the small or only the large predictions. The linear RMSE is 0.842947 tons. The yield formula adds noise with standard deviation 0.80, so most of the remaining error is that built-in noise, plus the farms whose fertilizer or moisture had to be imputed. The forest is less accurate here because the designed yield relationship is linear in the recorded inputs, apart from the absolute temperature term.



Held-out residuals for linear regression. Actual minus predicted, in tons.

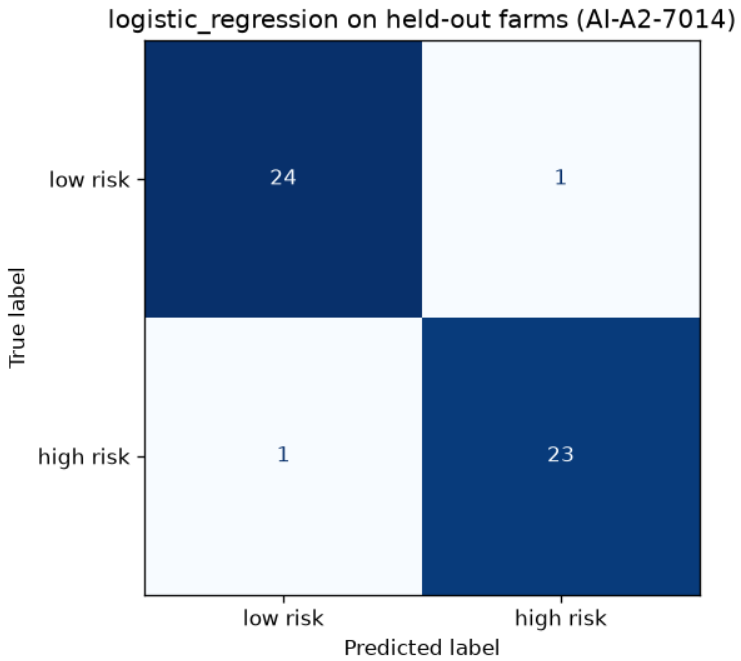
The largest absolute test error is row 241. Fertilizer is missing, so the model fills it with the training median of 136.2 kg. The farm has 3.3832 ha, 1316.3225 mm of rainfall, temperature 23.0246 C, season B, and cooperative support 0. Actual yield is 12.8769 tons and the prediction is 15.1372 tons, an overestimate of 2.2603 tons (artifacts/test_predictions.csv). Temperature is well above 18 C, and the formula penalises that gap with an absolute value, which a single straight line only approximates. Yield per hectare on the saved values is 3.8061, against a base-row median of 5.8521. The risk model still flags this farm, with probability 0.9887. The tonnage is the part a field officer should not treat as exact while fertilizer is blank.

High risk, held-out test set

Model	Precision	Recall	F1	ROC-AUC
majority_baseline	0.000000	0.000000	0.000000	0.500000
logistic_regression	0.958333	0.958333	0.958333	0.998333
random_forest	0.958333	0.958333	0.958333	0.996667

logistic_regression ties random_forest on held-out F1 for high_risk=1 (0.9583). It is selected because its ROC-AUC is higher (0.9983 versus 0.9967). The majority baseline F1 is 0.0000. These figures are the values stored in artifacts/classification_metrics.json.

Rows of the confusion matrix are the actual class and columns are the prediction, in the order low risk then high risk. Logistic regression has 24 true low-risk farms, 1 false alarm, 1 missed high-risk farm, and 23 true high-risk farms. The majority baseline predicts low risk for all 49 test farms, so its high-risk precision, recall, and F1 are 0. The forest makes the same two mistakes and ties logistic regression on F1. Logistic regression is kept because the ranking of probabilities is slightly better.



Confusion matrix for the selected logistic regression model.

The missed farm is row 41. All of its inputs are present: rainfall 1050.7080 mm, temperature 13.0229 C, soil moisture 54.2734 percent, fertilizer 91.1718 kg, field size 2.1062 ha, season B, and cooperative support 1. Actual yield is 11.7362 tons and the yield model predicts 11.8744 tons, an error of only -0.1382 tons. Yield per hectare is 5.5722, and the median on the saved base rows is 5.8521, so the farm sits just below the cutoff that defines high_risk. The high-risk probability is 0.3912, on the low-risk side of 0.5. I would not treat this as a wild miss. It is a borderline farm, and a person should look again when the probability is near one half.

5. Clustering

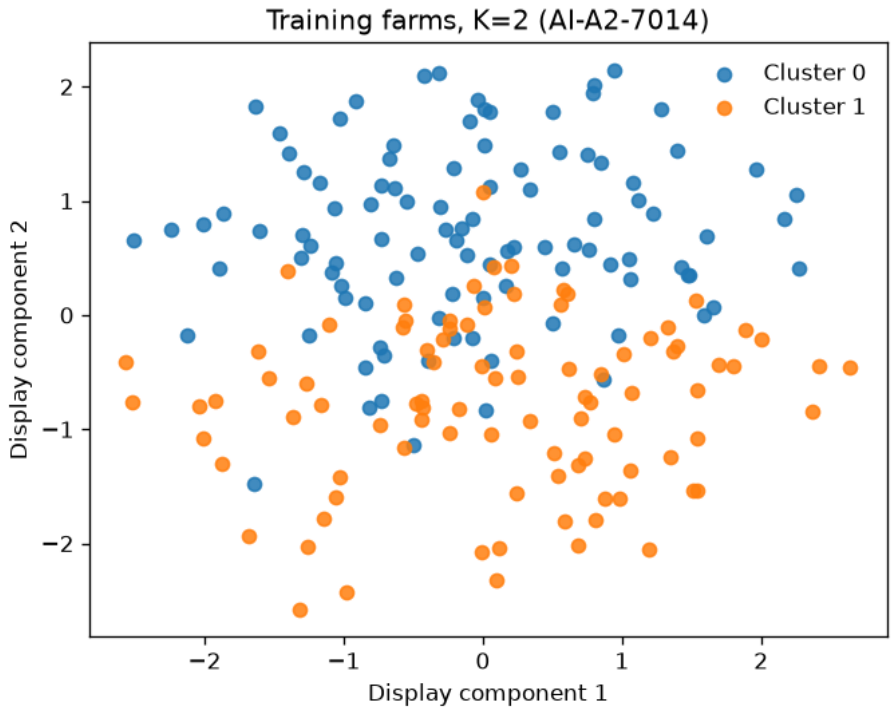
K	Silhouette on the training features
2	0.155880
3	0.140664
4	0.148118
5	0.140100
6	0.146105

K=2 has the highest silhouette score (0.1559) among K=2, 3, 4, 5, and 6. The score was calculated on training farms after imputation, scaling, and season encoding fit on the training split only. The separation is modest, so the groups overlap and should be treated as a rough organising aid.

Cluster 0 contains 102 training farms with more farms receiving cooperative support (100%). The group is a mix of season A and season B. Rainfall, temperature, soil moisture, fertilizer, and field size stay near the training average, so this group is defined by support status rather than by a different climate or field size. A field officer can use it to tell supported farms from unsupported farms. Yield and risk were not used to form the group.

Cluster 1 contains 92 training farms with fewer farms receiving cooperative support (0%). The group is a mix of season A and season B. Rainfall, temperature, soil moisture, fertilizer, and field size stay near the training average, so this group is defined by support status rather than by a different climate or field size. A field officer can use it to tell supported farms from unsupported farms. Yield and risk were not used to form the group.

Operationally, cluster 0 is the training farms with cooperative support and cluster 1 is the training farms without it. Rainfall, temperature, moisture, fertilizer, and field size stay near the training average in both groups, so I would not send different agronomic advice to the two groups. A field officer can use the label to organise contact with supported and unsupported farms. The silhouette is only 0.155880, and the display plot overlaps, so the groups are a rough aid. clusters.png is a two-component PCA of the scaled training features for display. Membership comes from KMeans on the full scaled feature set. Yield and high_risk were not used.



Training farms coloured by the selected KMeans labels. The axes are for display only.

6. API, testing, and model card

Endpoint contract

- GET /health returns status ok and the verification code AI-A2-7014.
- GET /model-info returns the student name, registration number, selected models, K, feature list, and the dataset SHA-256.
- POST /predict accepts the seven features and returns yield_tons, high_risk, high_risk_probability, cluster, and the verification code.

samples/valid.json is rainfall 1100, temperature 18.5, soil moisture 55, fertilizer 120, field size 1.5, season A, and cooperative support 1. The service returns yield_tons 12.3323, high_risk 0, high_risk_probability 0.0418, cluster 0, and verification code AI-A2-7014.

samples/invalid.json is the same farm with rainfall 50. Both predict.py and POST /predict reject it. The API returns HTTP 422 with field rainfall_mm and the message that the value must be between 600 and 1400. No score is returned. Omitting soil_moisture_pct is also a 422. Sending season as a number is a 422. The command-line tool exits with status 1 and prints the reason.

Tests

pytest -q collects six tests in tests/test_service.py, and all six passed when this report was built. test_valid_inference_returns_the_verification_code checks the command and the score fields. test_missing_field_is_rejected, test_wrong_type_is_rejected, and test_range_error_is_rejected each expect a rejection from the scorer and a 422 from the API. test_dataset_and_prediction_are_reproducible rebuilds the frame twice with seed 7014, checks the CSV hash against data_fingerprint.txt, and scores the same farm twice. test_api_health_model_info_and_predict calls /health, /model-info, and /predict.

Model card

Intended use: help a field officer, in a classroom setting, estimate harvest tons, notice farms the model marks as higher risk, and see which recorded farms share cooperative support. Prohibited use: credit, insurance, subsidies, public-service decisions, and any claim that the number is a real agronomic measurement.

Limits and risks: the file has 243 farms after the copies are removed, and only 49 are held out. The high F1 reflects a strong designed signal on a small test set, not a survey of Musanze. A blank fertilizer cell is filled with the training median, which is what went wrong for row 241. The clusters mostly rediscover a column that is already on the form. A field officer must review every estimate, and should look harder when an input is missing or the risk probability is near one half.

With real cooperative records I would replace the formula and the median risk rule with measured harvests and an agronomist's label, keep a validation split separate from the final test farms, and check whether cooperative support is a cause or only a flag already written on the sheet.